ANCIENT AGRICULTURE AS TECHNOLOGY

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THE PLAN

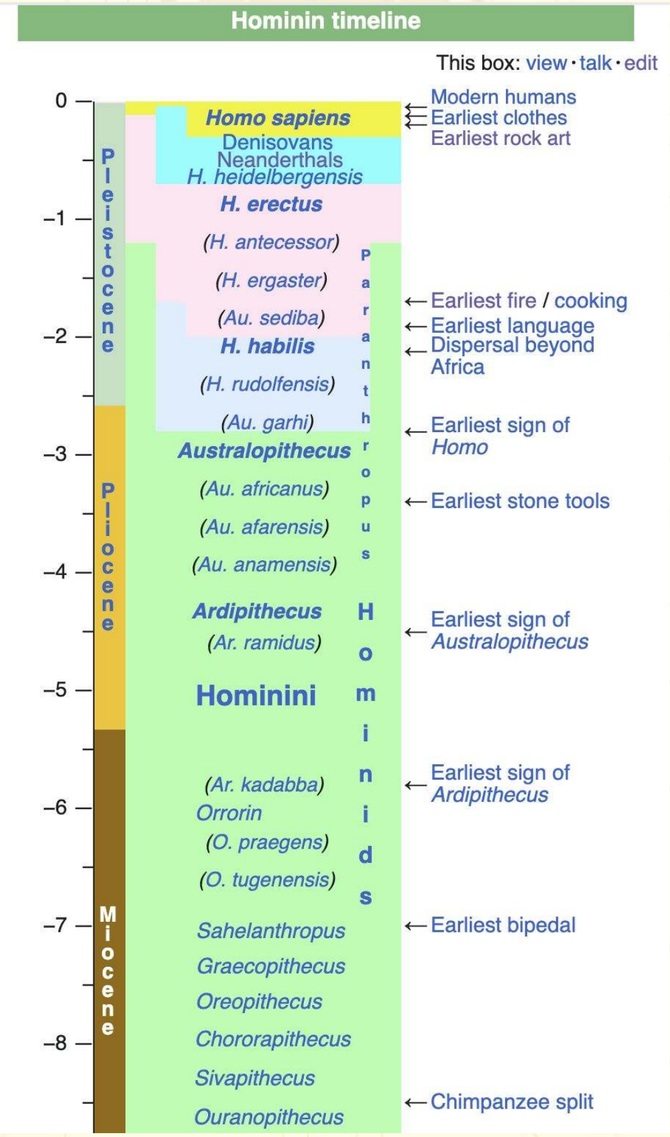
- Remember: Humans are Part of Nature
- Human “Niche Construction” and the Environment-Evolution Feedback Loop
- The Earliest Tools: Fire & Stone
- Agriculture as a Technological System
- Grain and the State

SOME TOOL USERS

- Tool use is *not* uniquely human.
- Chimps, bonobos, gorillas, dolphins, crows, otters all make and use tools.
- So *tool use* alone can't be the line that separates us from the rest of nature.

HUMAN EVOLUTION

- *Homo sapiens* – last remaining human species.
- Evolved from *H. erectus* ~300,000 YA (years ago).
- Other species of humans survived until ~30 KYA – ~40 KYA.
- Earliest *H. sapiens* inherit both stone tools & fire from pre-*sapiens* ancestors & contemporaries.



STONE TOOLS HAVE A HISTORY

- Earliest: “Oldowan Industry” (2.9–1.7 MYA).
- Next: Acheulian (1.95–0.13 MYA).
- Later: Mousterian (160–40 KYA).

TOOLS AND EVOLUTION

- Creation of tools: complex attention & dexterity.
- Use of tools: necessary for preparation of animal hides for clothing.
- Communication about & through tools: may have contributed to language development.

(Ambrose 2001; Morgan et al. 2015; Davidson 2011)

FIRE

> *"...it is no exaggeration to say that we are utterly dependent on fire. It has in a real sense domesticated us."*
>
> – *(Scott 2017, 42)*

FIRE (USES)

- Cooking
- Heat
- Landscaping

> *"...it is no exaggeration to say that we are utterly dependent on fire. It has in a real sense domesticated us."*
>
> *– Scott (p. 42)*

ECOLOGICAL NICHE

- Functional “locations” within an ecosystem.
- Match between organism’s capabilities & environment.

NICHE CONSTRUCTION

- In real ecosystems, organisms *construct* their niches.
- Dynamic relationship between traits & environment.

FIRE FOR NICHE CONSTRUCTION

- Fire clears forests, aids hunting.
- “Reduces radius of a meal” (Scott).
- Enables cold-weather expansion.
- Encourages cooperation, role division.
- Supports social interaction, language.

> *“a deliberate disturbance ecology in which hominids create… a mosaic of biodiversity and a distribution of resources more to their liking.”*
>
> *– (Scott 2017, 40)*

FROM FIRE TO AGRICULTURE

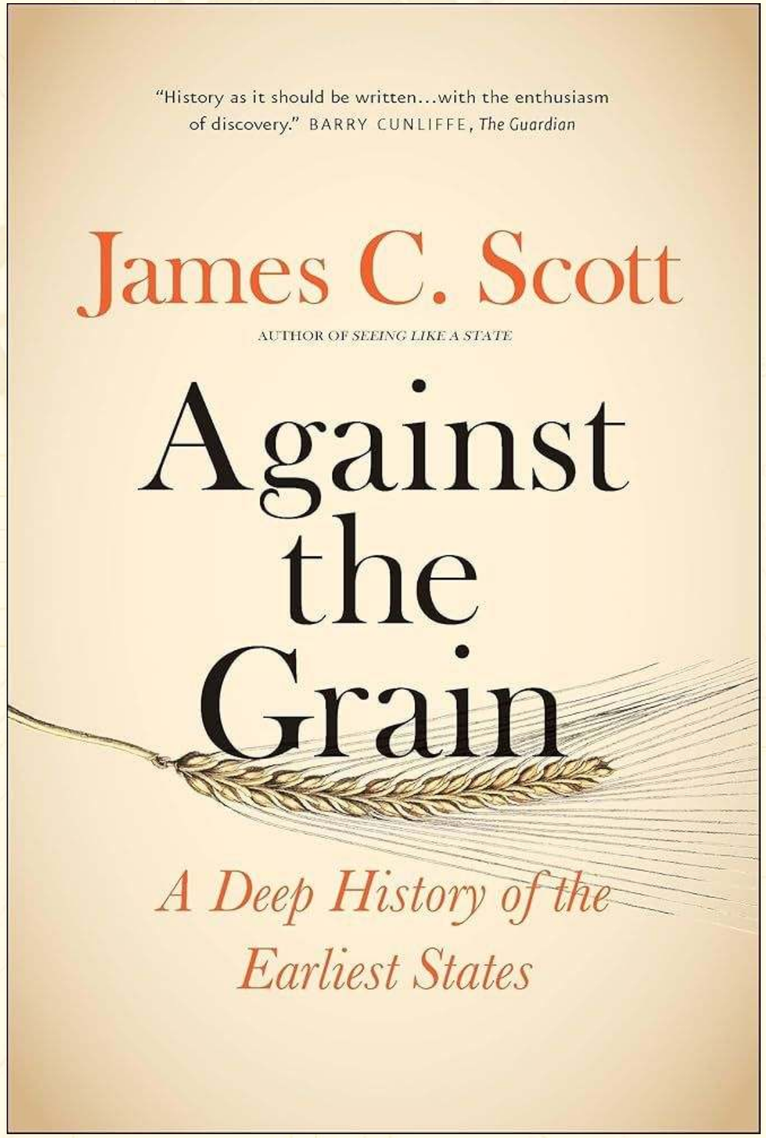
- Emerged ~12,000 years ago.
- ≥6 independent centres worldwide.
- Major: Fertile Crescent, Yangtze, Indus, Mesoamerica, Andes, New Guinea, West Africa.
- From foraging to sedentism.
- Early farming = first states.

WHY AGRICULTURE?

- Worse health, more labour per calorie.
- Population growth.
- Social hierarchy.

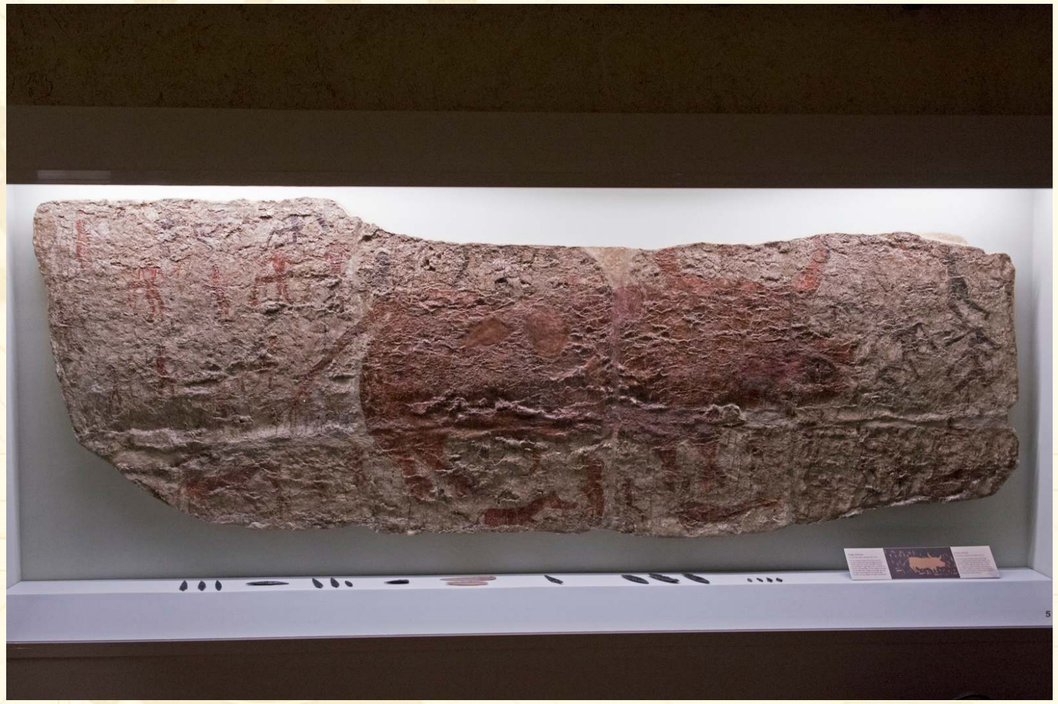
WHY AGRICULTURE?

- Agriculture = evolutionary, not sudden.
- Hunter-gatherers modified landscape.
- Used fire, planted seeds, built traps.



THE DOMUS: ANIMALS JOIN THE BARGAIN

- Plants weren't the only domesticates.
- Sheep, goats, cattle, pigs, dogs – domesticated 12,000–8,000 BCE.
- Plus *commensals*: rats, sparrows, lice – uninvited but inseparable.
- Consequences:
  - Caloric surplus (dairy, traction, meat)
  - Zoonotic disease (smallpox, measles, flu)
  - The first epidemics



Wall painting of a bull surrounded by hunters, Çatalhöyük (Anatolian Neolithic, c. 7000–6000 BCE), Museum of Anatolian Civilizations, Ankara. Photo by Dosseman, CC BY-SA 4.0.

AGRICULTURE AS A SYSTEM

- Recall Ede: technology as *the system* for solving real-world problems.
- Agriculture is the largest such system humanity had built so far.
- Components:
  - Seeds & soil (selected, modified, exhausted)
  - Tools: plow, sickle, threshing floor, granary
  - Labour: organized, gendered, eventually coerced
  - Knowledge: when to plant, when to harvest, how to store
  - Records: marks, tokens, eventually writing

Scribes recording the harvest, Tomb of Menna (Egyptian, c. 1400 BCE), West Thebes. Photo by Soutekh67, CC BY-SA 4.0.

THE PROPERTIES OF GRAINS

- Large seeds, stable to collect.
- Grow above ground, visible.
- Harvested all at once.
- “The tax-man’s harvest.”

A FIRST TAX RECEIPT

Late Uruk-period clay tablet (c. 3300–3100 BCE) recording quantities of barley and emmer wheat. Photo by Osama Shukir Muhammed Amin FRCP(Glasg), CC BY-SA 4.0.

- Records who got how much barley, in what container.
- Among the earliest writing in the world.
- Writing began as bookkeeping, not poetry.
- Grain → record → tax → state.

GRAINS AND STATES

> *"What is required is wealth in the form of an appropriable, measurable, dominant grain crop and a population growing it that can be easily administered and mobilized." (p. 24)*

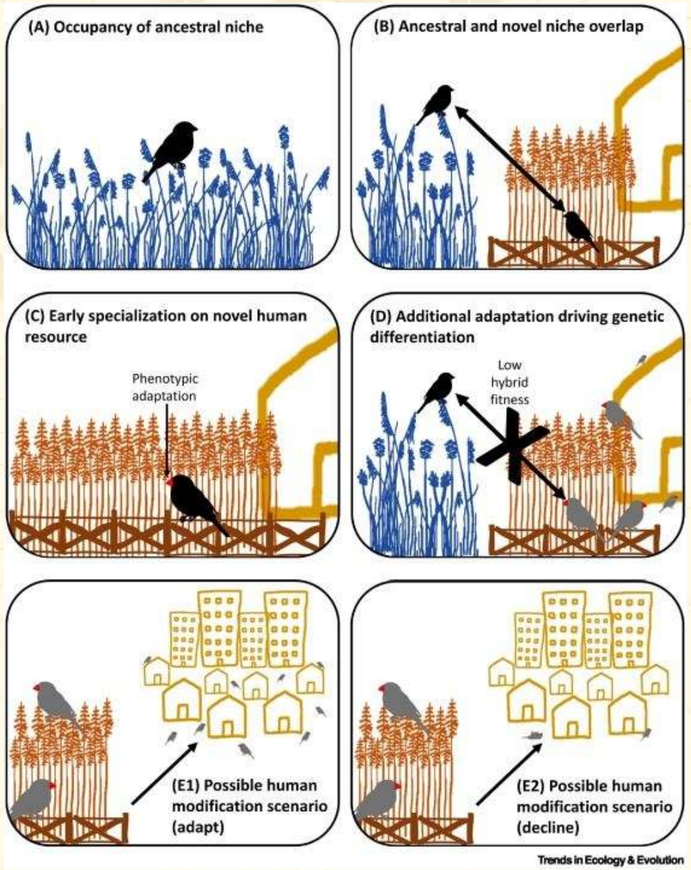
(Scott 2017)

GRAINS AND STATES (INNOVATION IN SERVICE OF THE STATE)

- Slow cycle of innovation *in the service of state formation*.
- Importance of controlled environments; artifacts such as the plow make this possible.
- Dozens or hundreds of other innovations.

DOMESTICATION & THE ANTHROPOCENE

- “Thin” Anthropocene begins early, with fire & agriculture.
- Farming complex w/ domesticated animals & semi-domesticated “commensals” radically transforms our “niche.”
- Creates drastic new social forms (perhaps misleadingly called “progress”).
- Also domesticates us!



TAKE-AWAYS

- Farming is not the opposite of technology – it is one of our oldest and most consequential technological systems.
- Agriculture builds on deep-time practices: fire, niche construction, tool-use.
- It is a *system*: plants + animals + tools + labour + records + state – and each part pulled the others into being.
- Grain made the state possible; the state then reshaped the grain.
- More next time!

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